

MACHINE LEARNING / DEEP LEARNING

Machine Learning (ML)

Enable Systems to learn from data using algorithms and statistical model

Approach
Uses Traditional algorithms to learn Patterns from data

Feature Engineering
Manual Feature extraction by domain Expert

Data Requirement
Works well with smaller dataset

Training Time
Generally, fast training

Computing Power
Requires less Computational and Explainable

Interpretability
More interpretable and Explainable

Best Suited For
Structured data (eg. tables, records, numeric data)

Algorithms Used.
Linear Regression, logistic regression, SVM, Decision Tree, Random Forest, KNN, Naive Bayes...

Both are subset of

Artificial Intelligence..

Enables Computers to learn from data and make decisions



Key Comparisons

Machine Learning	Aspect	Deep Learning
Subset of AI	Relationship	Subset of ML
Manual Feature Extraction	Feature learning	Automatic Feature learning
Less Data	Data Requirement	Large Data
Faster Training	Training Time	Slower Training
Lower Computing Power	Computing Power	Higher Computing Power
More Interpretable	Interpretability	Less Interpretable
Works well on Structured Data	Data Type	Works well on unstructured data

Real World Application of Machine & Deep Learning

Deep Learning (DL)

Uses artificial neural network with many layers to learn representation from data

Approach
Uses multi-layer neural networks to learn patterns

Feature Engineering
Automatic feature extraction by the model

Data Requirement
Requires large dataset for better Performance

Training Time
Generally longer training time

Computing Power
Requires high Computational Power (GPU/TPU)

Interpretability
Less interpretable (black box)

Best Suited For
Unstructured data (eg. images, audio, video, text)

Models used
ANN, CNN, RNN, Transformers

- Spam Detection
- Credit Scoring
- Medical Diagnosis
- Recommendation Systems
- Fraud Detection
- Speech Recognition
- Image Classification

1. Personalized Learning

Creates Customized learning paths
Adapts to individual student needs and pace
Recommends resources and practice exercises

2. Intelligent Tutoring & Support

AI tutors and chatbots available 24x7
Explains concepts in simple language
Helps in doubt clarification and step by step solution

Language and Writing Assistance

Improves grammar, clarity and style
Assists in essay writing, reports and assignment
Translates languages and paragraphs content

Content Creation

Generate notes, summaries, articles and explanations
Create lesson plans, PPTs and study material
Produce images, videos, diagrams and simulations

APPLICATIONS of Generative - AI (Education)

Creativity and Innovation

Generates ideas and brainstorm new concepts
Helps in project planning and design thinking
Encourages creative writing, storytelling & coding

Transforming Education

Generative AI empowers teachers & students to learn smarter, teach better and innovative for a bright future....

Assessments and Feedback

Generate Quizzes, Questions and practice
Evaluate answers and assignment
Provides instant feedback and suggestions

Research & Knowledge Discovery

Summarize large research papers and articles
Find relevant information and reference
Support literature review and data interpretation

Accessibility & Inclusion

Converts text to speech and speech to text
Simplifies complex topics
Supports learners with disabilities & diverse

Benefits

Saves Time
Improves Grades
Better Learning
Inspires Creativity

Challenges

Ethics & Bias
Privacy & Data Security
Academic Integrity
Over-reliance on AI
Need for Digital Literacy